

THERMODYNAMIC GRAMMAR

Systems, energy transfer and the greenhouse effect | 25 minutes | 30 marks | no calculator

Name _____

Date _____

Instruction: make every claim carry its physics. Name the system, the transfer or interaction, the quantities and the relationship. A familiar phrase earns no credit if it hides the mechanism.

1 | DESCRIPTION, PSEUDO-EXPLANATION OR EXPLANATION

6 MARKS • 5 MIN

Write **D** for direct description or measurement, **P** for an explanation-shaped restatement or label, and **E** for a model-based explanation linking a process or relationship to evidence.

CODE	STATEMENT
☐	The gas temperature rose from 295 K to 318 K while its volume remained constant.
☐	The gas became hotter because heat made it warmer.
☐	Energy transferred through the cylinder wall increased the gas's internal energy; at fixed volume, no boundary work was done, so its temperature rose.
☐	Satellite data show 238 W m^{-2} absorbed as short-wave radiation and 235 W m^{-2} emitted to space as long-wave radiation.
☐	The planet warmed because the greenhouse effect warmed it.
☐	With outgoing energy transfer below absorbed energy transfer, the Earth system gains energy and warms until outgoing long-wave emission again matches the absorbed rate.

2 | EQUATION AS GRAMMAR: SYSTEM, STATE AND PROCESS

8 MARKS • 7 MIN

Situation: An ideal gas in a sealed, thermally insulated cylinder is compressed slowly by a frictionless piston. For this item use $\Delta U = Q + W_{\text{on}}$, where positive Q and positive W_{on} transfer energy into the gas.

Define the system and its boundary. State whether matter crosses that boundary. [2]

State the value of Q and the sign of W_{on} . Justify each from the situation. [2]

Explain why U is a state quantity while Q and W_{on} are process quantities. [2]

Give a microscopic explanation for the temperature rise during the compression. [1]

State one ideal-gas assumption and one condition under which it becomes less adequate. [1]

Checkpoint: insulation constrains one transfer pathway. It does not mean that the gas cannot gain energy.

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3 | RADIATIVE BALANCE: QUANTITIES, STREAMS AND CLOSURE

8 MARKS • 7 MIN

One-box Earth model: mean incident solar flux at the top of the atmosphere = 340 W m^{-2} ; planetary albedo $\alpha = 0.30$. Just after greenhouse-gas concentration rises, outgoing long-wave flux is 235 W m^{-2} . Assume reflected short-wave flux is unchanged.

Define albedo as a relationship between two radiative quantities. [1]

Calculate the absorbed solar flux. [1]

For both the incident solar stream and the outgoing terrestrial stream, state direction and wavelength range. [2]

Calculate the net radiative flux into the Earth system and state what its sign means. [1]

Explain the subsequent warming and new steady state using absorption or emission and the flux imbalance. Do not write 'heat is trapped'. [2]

State one limitation of treating Earth as one uniform box. [1]

4 | LANGUAGE FORENSICS

6 MARKS • 4 MIN

Match each sentence to one error family. Use each once: **substance** | **agent** | **attribute** | **process-state** | **model-object**.

ERROR FAMILY	SENTENCE
_____	A. The atmosphere stores heat like a blanket.
_____	B. Entropy pushes every system towards disorder.
_____	C. The gas contains more temperature after compression.
_____	D. At radiative equilibrium, radiation stops flowing.
_____	E. The one-layer atmosphere in the diagram is the real atmosphere.

Choose one sentence and repair it with a physically precise relationship. [1]

5 | EXIT WARRANT

2 MARKS • 2 MIN

A student says: '**Absorbed and outgoing fluxes are equal, so the greenhouse mechanism has stopped.**'

What does equality of these two rates legitimately establish? [1]

Rewrite the claim to describe the continuing processes at steady state. [1]